Hallucination Machine 2.0: Using Substitutional Reality to Investigate both Superstitious Perception and the Impact of Sense of Presence on Reaction Times

Benjamin Oliver Barnett

Supervised by Dr. Keisuke Suzuki

Intelligent and Adaptive Systems MSc

UNIVERSITY OF SUSSEX

Abstract

VR has been used to elucidate aspects of cognition ranging from embodiment to psychedelic experiences. In a recent paper, Suzuki et al. (2017) introduced the 'Hallucination Machine': a VR set up that mimics psychedelic experiences in an immersive manner. However, the original set up failed to evoke a strong sense of presence in participants, calling their conclusions into question. The ability to increase presence in virtual environments is thus an important target for VR research, alongside deducing whether increased presence evokes cognitive changes. In order to explore whether Substitutional Reality (deceiving participants into believing a pre-recorded video is a live scene) may improve feelings of presence, we implemented an updated version of the Hallucination Machine and analysed performance in two behavioural tasks. The tasks used were a superstitious perception task and a change detection task, and we compared results across two groups of participants: those who underwent SR and those who were informed they were watching pre-recorded videos. Furthermore, we correlated scores of vividness of visual imagery with reaction times in the change detection task. We found the Hallucination Machine had no effect on participants' superstitious perception, regardless of condition. Similarly, no difference was found during performance of the change detection task. Vividness of visual imagery was, however, found to be a significant predictor of reaction times. Despite negative results, we have presented a methodology with which to discover ways of improving feelings of presence in VR systems. Indeed, it is argued that this is a crucial line of investigation in order to instill a sense of presence closer to that of 'normal' consciousness.

1 Acknowledgements

I am beyond grateful for the help gifted to me by Dr. Keisuke Suzuki throughout the course of this project. His assistance in integrating the Deep Dream Videos with Unity in order to use them with VR hardware enabled the experiment to be set up within a realistic time frame, and his advice and suggestions throughout all stages of the process were both constructive and encouraging.

2 Introduction

2.1 Substitutional Reality Vs. Virtual Reality

The use of virtual reality (VR) within cognitive science has increased steadily over the past decade. By enabling researchers to design their own virtual environments (VEs), VR allows for the systematic and parameterizable manipulation of perceptual experiences (Suzuki et al., 2017) alongside the analysis of the resultant cognitive effects. Despite the rise in VR research, the difference between VEs and physical reality are still vast, such that the conclusions drawn from VR studies may not apply to perception of the natural world. For instance, the 'Hallucination Machine' (Suzuki et al., 2017), a combination of machine learning and immersive VR, was designed in order to elucidate various aspects of the psychedelic experience. Specifically, Suzuki et al. aimed to dissociate the systemic effects of psychedelic compounds (i.e. global physiological changes) from consequences of manipulated of sensory processing, such as altered visual sensations. By creating a VE that provided a realistic replication of the visual aspect of psychedelic experience, the authors were able to investigate whether the temporal distortion commonly reported in altered states of consciousness (ASC; Boardman et al., 1957) was a product of pharmacological effects of psychedelic

compounds or whether it emerged due to a disrupted and vastly altered visual experience. Since time perception was found to be unaltered within the hallucination machine, it was concluded that temporal distortion in the psychedelic state must be due to general systemic effects of psychedelic compounds (Suzuki et al., 2017).

Despite results from the ASC suggesting the hallucination machine gave rise to subjective experience similar in multiple dimensions to the psychedelic state, participants commonly reported that the experience was little more than 'watching a video' (Suzuki, personal communication). Such a report is indicative of a lack of a feeling of presence. The feeling of presence can be defined as the subjective sense of reality of the world and of the self within the world (Metzinger, 2013). In VR research in particular, presence is used to describe the sense of being in a VE rather than the physical environment (Sanches-Vives and Slater, 2005). It is perhaps not surprising that Suzuki et al.'s set up failed to induce a strong sense of presence, since the Hallucination Machine consisted of visual stimuli pertaining to an outside area, whilst all other modalities were consistent with being inside a psychology lab. Indeed, multi-modal, congruent cues enhance the sense of presence (Sanches-Vives and Slater, 2005; Dinh et al., 1999), underscoring how a visual-only VE is unlikely to provoke strong feelings of presence in participants.

Evoking a sense of presence is imperative, since true immersion in a VE facilitates a number of cognitive changes that are otherwise much weaker or absent altogether. For example, a number of studies have shown that a heightened sense of presence within a VE improves performance on subsequent memory tasks relating to the virtual world (Dinh et al., 1999; Lin et al., 2002; Mania and Chalmers, 2001; c.f. Fox et al., 2009, Bailey, 2012). Moreover, the sense of presence has also been found to be a significant predictor of emotional variables (Riva, 2007), underscoring its importance to both cognitive and affective domains. As such, conclusions drawn from studies such as Suzuki et al. (2017) should be treated with caution; during non-pathological perception of the natural world, a sense of presence is consistently present, such that it is a 'structural component of consciousness' (Seth et al., 2012). As such, the absence of presence in the VE used by Suzuki et al. casts doubt over their conclusions, since they were investigating a state of mind that was dissimilar and less present from the consciousness they formed conclusions about.

A successful method of maintaining a sense of presence within a VE is the use of Substitutional Reality (SR) techniques (Suzuki et al., 2012). This methodology covertly substitutes participants' live reality with a pre-recorded reality without them being aware of the change. An SR session begins by showing participants a live scene through a Head-Mounted Display (HMD), inducing a strong sense of presence within the SR system that continues after the recorded scene replaces the live feed (Suzuki et al., 2012). Indeed, the system is so compelling that even after participants have been informed of how the SR functions, they are still unable to accurately differentiate live

from pre-recorded scenes (Suzuki et al., 2012). In line with this, the primary aim of the present study is to compare the cognitive effects of utilising a SR methodology compared to a standard VR methodology, where participants are aware they are watching pre-recorded videos. Using an adapted version of the original Hallucination Machine, we predicted that participants who believe they are watching a live feed (SR condition) will have an increased sense of presence compared to those who do not (NONSR). In line with this, we also hypothesised that participants in the SR condition would display quicker reaction times in a simple detection task than those in the Non-SR condition, since the aforementioned studies have alluded to perception that is more detailed during heightened states of presence (Lin et al., 2002; Mania and Chalmers, 2001). As such, the present study aims, in part, to highlight how the induction of a sense of presence in VR experiments is vital in order to draw valid conclusions about natural perception by illustrating a difference in performance in cognitive tasks as a function of feelings of presence.

2.2 Superstitious Perception

Alongside our simple detection task, this study also utilises the superstitious perception paradigm (Gossellin, 2003) to investigate the modulation of top-down vs. bottom-up information within a predictive coding framework. Briefly, predictive coding describes the existence of a neuronal hierarchy, whereby higher neural regions generate predictions about representations in lower levels (Clarke, 2013). These top-down predictions are subsequently compared with the actual representations within lower levels, generating prediction errors. Prediction errors are then passed back up the hierarchy, such that the predictive model can be updated; permitting more accurate predictions in the future. Ultimately, the cyclical exchange of signals across layers minimises prediction error at each layer, resulting in a deep, neurally encoded 'explanation' for sensory input (Seth, 2018).

The balance of top-down and bottom-up information within this system (i.e. how much the brain relies on either prior expectations or raw sensory data during perception) is controlled by precision weighting. Every prediction error has its own precision, which signifies the 'confidence' the system has in either sensory input or stored priors (Seth, 2013). As such, precisions control the level to which prediction errors are resolved, such that optimising precision weightings causes the relative influence of prior expectations to be balanced against that of the prediction errors, which are directly related to the bottom-up sensory input. Specifically, low precision on error signals denotes higher reliance on top-down prior beliefs during perception. The effect of precision weighting in the balance of top-down and bottom-up information is illustrated in Figure One.

This system of balancing top-down and bottom-up information during perception is posited to underlie the positive symptoms of schizophrenia, namely hallucinations and delusions (Fletcher and Frith, 2009). By applying pathologically high precision weightings to top-down knowledge,

Figure 1: Replicated from Seth (2013). The curves are probability distributions over the value of a particular sensory stimulus (x-axis). *Left*: The sensory signals (red) have a high precision weighting, which enhances their influence on the belief about the value of the sensory stimulus (the posterior; green) as compared to the top-down prior (blue). *Right*: when the sensory signals have a low precision weighting, there is an opposite effect on the posterior.

individuals suffering from such disorders are likely to experience false perceptions based on their internal representations. In other words, patients undergo perceptions of objects or events that do not exist in their bottom-up sensory information.

The deep dream network and process has a number of similarities to this mechanism of biological hallucination. For instance, there is a close mapping of the DCNN architecture to that of the primate visual system (Kriegeskorte, 2015), illustrated by the fact that features detected by lower layers within a DCNN closely match those low-level features processed in early parts of the visual cortex, such as V1. Moreover, the neural responses of the Inferior Temporal cortex, an area strongly associated with object recognition, have been shown to resemble the activity patterns of higher layers of DCNNs when activated by visual stimuli (Caideu et al., 2014). Moreover, the deep dream process relies on exaggerating the high-level features stored within the network and imposing them on the input, such that resulting images are embedded with what can be seen as the 'top-down' representations of the network (see Section 2.2.4). This bears a striking similarity to the predictive coding account of hallucinations, inasmuch as higher weights are placed on top-down processing as opposed to bottom-up input, modifying perception such that what is perceived (outputted) is not actually occurring within the 'real world' (input image). Because of this congruence between the deep dream algorithm and biological hallucinations, the use of the deep dream process allows for the production of naturalistic, yet unreliable, visual stimulation.

The optimisation of precision weighting based on the reliability of sensory input has been explicated in a number of scenarios (Ernst and Banks, 2002; Stocker and Simoncelli, 2006; Behrens et al., 2007; Kording and Wolpert, 2004). For instance, by varying the noise associated with sensory data whilst asking participants to compare the height of two ridges by touch and vision independently, subjects' behaviour during subsequent multimodal experiments could be predicted (Ernst and Banks, 2002). This was done by calculating the visual and haptic discrimination thresholds during the unimodal trials and applying these values to a Maximum Likelihood-Estimator (MLE) integrator. Indeed, participants displayed behaviour that was predicted by the MLE integrator, suggesting they were integrating the information optimally. Specifically, they behaved in a way

concurrent with the notion of 'visual capture', whereby vision contributes more than haptics in bimodal tasks. Moreover, for any system implementing MLE style optimisation, visual capture will occur whenever the precision of visual stimuli is greater than the haptic stimuli (Ernst and Banks, 2002).

More relevant to the present study, however, is the balance of prior expectations against bottom-up input, rather than optimising input from two different modalities. An illustrative example of this phenomenon is offered in the case of a particular motion illusion, whereby low contrast stimuli are perceived to move more slowly than those of high contrast, even when their velocities are equal (Ernst, 2010). It has been posited that this may be due to a prior expectation that objects will move slowly, given that objects tend to only move at low speeds or not at all (Weiss et al., 2002). When sensory stimulation is noisier, this prior will receive a higher precision and thus more weight during perception. Indeed, low contrast stimuli is less reliable in providing motion signals than high contrast stimuli, resulting in them being perceived to have lower velocity. This hypothesis was confirmed by reverse engineering the shape of the prior distribution and likelihood function from participants' perceptual behaviour and fitting a Bayesian estimator model to trial by trial responses during a speed discrimination task (Stocker and Simoncelli, 2006). In line with Weiss' (2002) hypothesis, the prior distribution associated with the most successful model, which accurately predicted participants' behaviour, was that of a low velocity. Alternative research has *created* the prior expectations within the experimental paradigm (rather than deducing the prior theoretically) and observed how participants will then behave in a way consistent with perceiving the prior when they are uncertain about the sensory input (Kording and Wolpert, 2004).

Overall, the increased reliance on prior expectations when observing unreliable sensory input is well documented. However, it is yet to be shown whether unreliable, yet naturalistic, sensory stimulation will produce similar results. Due to the complexity of natural images, the extent to which studies based on artificial images elucidate facts about natural vision is far from clear (Yuille and Kersten, 2006). For example, responses of neurons in V1 to natural images cannot be predicted by their response to artificial stimuli (Carandini, 2005). Because of this, it is important to replicate previous observation of precision optimisation with natural images, rather than the noisy artificial images used in previous research in precision weighting (Ernst and Banks, 2002; Behrens et al., 2007; Kording and Wolpert, 2004). Indeed, the sensory input utilised within the Hallucination Machine are consistent with reports of psychedelic experiences (Suzuki, 2017), and, in combination with the aforementioned similarities between biological hallucinations and the deep dream algorithm, are therefore highly naturalistic. We thus aimed to illustrate that reducing the reliability of bottom-up sensations in a naturalistic manner can cause an optimisation of precision weightings such that prior expectations will be relied on more heavily.

The superstitious perception paradigm offers a technique with which to reconstruct internal representations of an observer (Gosselin, 2003). This paradigm consists of viewing a number of white noise images, within which participants have been instructed there are specific patterns, despite there being none. Since white noise has no coherent structure, the perception of any such patterns must arise from the observers' internal representations. The superstitious perception paradigm has been used to visualise the internal representations of the letter 'S' in three participants (Gosselin, 2003), by combining all the stimuli that were deemed to contain an 'S' and those that were rejected, and then subtracting the 'No' image from the 'Yes' image. It was then shown that if the white noise image correlated even slightly with the superstitious observers' internal representation of an 'S', they were far more likely to respond positively to the image, underscoring the influence of top-down priors during perception.

Superstitious perception has also been shown to be affected following cognitive manipulations (Whitson, 2008). When participants experienced a lack of control, caused by erroneous feedback about their actions, illusory pattern perception was found to increase, supposedly because pattern perception is a 'compensatory mechanism designed to restore feelings of control' (Whitson, 2008).

Taking the aforementioned results together, we hypothesised that reducing the reliability of visual input by using the Hallucination Machine would increase the precision weighting of top-down priors in comparison to bottom-up sensory input. As such, we predicted that participants would see more illusory patterns within white noise images at the end of their immersion in the Hallucination Machine, rather than at the beginning, since they would be relying on their internal representations more than their visual input.

2.3 Vividness of Visual Imagery

The final aim of this study was to investigate individual differences in vividness of visual imagery and to examine whether these differences correlated with performance on the detection task. The Vividness of Visual Imagery Questionnaire (VVIQ) (Marks, 1973) has been shown to be a useful measure of vividness of visual imagery (McKelvie, 1995). However a variety of analyses and tasks have failed to show a relationship between this subjective measure of vividness of visual imagery and objective behavioural measures that, theoretically, should be influenced by vividness of visual imagery (Ernest, 1977; McKelvie and Rohberg; 1978; Richardson, 1980). Whilst it has been suggested that the VVIQ primarily measures memory imagery (Hilgard, 1981), it is certainly conceivable that those with vivid mental imagery will take longer to detect hallucinatory patterns, given that presumably participants are somewhat habituated to the existence of patterns during visual experience. As such, we hypothesised that higher ratings of vividness, as measured by the VVIQ, would positively correlate with reaction times in the detection task.

To summarise our results, there was no difference in subjective measures of presence between groups. Moreover, there was no difference in reaction times across conditions during the change detection task. Secondly, there was no effect of the Hallucination Machine on either the SR or the NONSR group's superstitious perception. Finally, we were able to show that participants' VVIQ scores were significant predictors of their reaction times during the detection task.

3 Methods

3.1 Real-Time Deep Dream Approach

We initially aimed to produce a real-time 'Hallucination Machine', whereby video input from a camera placed on the front of the Head Mounted Display (HMD) would be converted into a deep dream style similar to that produced in Suzuki et al. (2017), without any perceptible delay.

In order to achieve this, a recent system that transfers the style of one image to another in real time (∼30ms; Johnson et al., 2016) was utilised. As illustrated in Figure One, this style transfer system consists of two networks, an Image Transormation Network fw and a Loss Network ϕ. The Image Transformation Network is a deep convolutional neural network (DCNN), which transforms an input image x into output images $\hat{y}$ via the mapping $\hat{y} = fw(x)$. This network is trained using gradient descent, minimizing a weighted combination of loss functions defined in the Loss Network. Each loss function within the Loss Network calculates a value that measures some difference between the output image and either a style target y_s or content target y_c. It is worth noting that the content target is by default the unaltered input image. The loss functions are described in detail below.

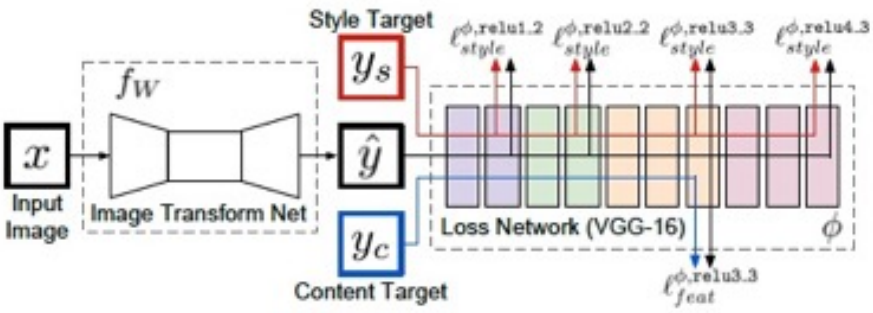

Figure 2: A Real-Time Style Transfer System, replicated from Johnson et al. (2016). The Loss Network defines multiple loss functions, specifically feature reconstruction and style reconstruction losses. These losses are used to tune the weights of the Image Transformation Network, which applies the style transfer to an input image.

3.1.1 Feature Reconstruction Loss

The Feature Reconstruction Loss forces the output image $\hat{y}$ to have similar high-level features to the content target image (i.e. the input image), as compared by the loss network. This prevents the two images from having perfectly matching pixel patterns, whilst still sharing the same feature representations. Specifically, the Feature Reconstruction Loss is the Euclidean distance between the feature representations of the input image and the output image, where $C_j H_j W j$ is the feature map of convolutional layer ϕ_j:

$$\mathcal{L}_{feat}^{\phi,j} = \frac{1}{C_j H_j W j} ||\phi_j(\hat{y}) - \phi_j(y)||_2^2$$

3.1.2 Style Reconstruction Loss

The Style Reconstruction Loss involves calculating a Gram Matrix for the output of the various layers of the loss network. The Gram Matrix (a matrix comprising of multiple correlated features) is simply a matrix of dot products of feature activations of a Loss Network layer. To clarify, if an entry in the gram matrix is close to zero, those two features in the given layer do not activate simultaneously for the given style. Similarly, if the value is large, the two features activate concurrently for images of that style. These activations patterns can then be replicated to produce images of similar styles to the style target y_s.

If the feature map of a certain layer in the loss network is matrix F, each entry in Gram Matrix G is given by:

$$G_{ij} = \sum_k F_{ik} F_{jk}$$

And the loss function is simply the mean squared error of the gram matrices when compared to the style target's gram matrices:

$$L_{style} = \frac{1}{2} \sum_{l=0} (G_{ij}^l - A_{ij}^l)^2$$

3.1.3 Initial Approach

A TensorFlow implementation of Johnson et al.'s (2016) system (https://github.com/hwalsuklee/) was adapted with the aim of producing deep dream like images. We hypothesised that by solely training the image transformation using the feature reconstruction loss on a high layer of the network, we could produce images whereby image content is preserved but lower level features such as colour, texture and boundaries are not. In this way, we hoped to produce images that

would replicate images produced via running deep dream algorithms on relatively low layers of a network. As such, the image transformation network was trained on layer Relu5_2 of the VGG16 Loss Network.

3.1.4 Results

The output produced by this process is presented in Figure Three. Training the network solely on a single feature reconstruction loss function failed to alter the content of the image, manipulating only the luminance of the input images.

Figure 3: *Left*: Input Images. *Right*: Output from the first attempt at producing deep dream like images from a real-time style transfer network. After training the image transformation network with only a feature reconstruction loss, images were altered only slightly.

3.1.5 Secondary Approach

Due to the failure of our initial approach, both the feature and style reconstruction losses were utilised in our subsequent attempts to produce real time deep dream images. We hypothesised that using style reconstruction loss on several high layers, whilst using feature reconstruction on only the lowest and highest layer would produce the pareidolia-like features seen in traditional deep dream images, when using an appropriate style target. This was predicted to be the best combination of layers to produce such images, since the high and low feature reconstruction loss would allow for high level features to be maintained from the input image, whilst maintaining the existing edges and boundaries within the image. The high-level style loss, however, was predicted to transfer the high level styles of the style target, for example a face of a dog in a picture filled with dog faces, onto the input image. The Style Target and Content Target used in this process are illustrated in Figure Four. Results from this process are presented in Figure Five.

The evident failure of this approach was concluded to be due to the inability of the system to

recreate high-level enough features and transfer them to the input image. In order for pareidolia-like effects, entire dog faces should be visible in the output image, yet only basic eyes were visible in some of the output images. Thus, in order to ensure convincing hallucinatory stimuli, we abandoned the real-time deep dream approach in favour of a pre-recorded videos

Figure 4: *Left*: The Style Target. *Right*: The Content Target.

Figure 5: Output following experimentation with multiple combinations of network layers. *Left*: Using layers Relu1_2 & Relu5_1 for Content Loss (weight = 7.5) and layers [Relu1_1, Relu2_1, Relu3_1, Relu4_1, Relu5_1] for Style Loss (weight = 5×10^2). *Center*: Using layers Relu1_2 & Relu5_1 for Content Loss (weight = 50) and layers [Relu4_1, Relu4_2, Relu5_1, Relu5_2] for Style Loss (weight = 5×10^2) . *Right*: Using layers Relu1_2 & Relu5_1 for Content Loss (weight = 3×10^2) and layers [Relu4_1, Relu4_2, Relu5_1, Relu5_2] for Style Loss (weight = 5×10^2).

3.2 Materials

3.2.1 Hallucination Machine

In order to create the Hallucination Machine, the deep dream algorithm was applied to every frame in a 30 second, 360 degree prerecorded video. The subsequent 'deep dreamed' video was then presented to participants in a Head Mounted Display (HMD). Participants were able to freely explore the virtual environment by moving their head.

3.2.2 Video and Presentation

The video footage was recorded in the Virtual Reality Lab at the University of Sussex using a 360 Degree video camera (Richoh Theta S 360°). The frame rate of the video was 29 fps at a resolution of 3840 x 1920. The HMD used to display the footage was an Oculus Rift, Development Kit 2, supported with software developed in-house, using Unity3D.

3.2.3 Deep Convolutional Neural Network (DCNN)

The DCNN used to produce deep dream videos, GoogleNet, was previously trained to classify images into one of 1000 categories in the ImageNet Large-Scale Visual Recognition Challenge 2014 (ILSVRC 2014; Szegedy, 2015). Backpropagation is used in this training process to modify the weights of the network such that the network assigns training images to specific categories. Eventually, the trained network is able to implement a direct mapping from an image's pixels to the categories, resulting in an error of only 6.67% in the ILSVRC 2014, with the difference between categories often incredibly subtle, such as that between the Siberian Husky and Eskimo Dog (Szegedy, 2015). This GoogleNet network was used to create the deep dream videos we utilised in this study.

3.2.4 Deep Dream

The deep dream algorithm uses a modified backpropagation process to produce hallucinatory-style images. Instead of updating the weights to minimize error, it fixes the weights across pre-defined layers of the network, iteratively altering the input image such that categorisation error is reduced. Following multiple iterations of this process, the input image will be modified such that it displays features learnt by the DCNN during training (e.g. a snail). When fixing a lower DCNN layer, the resultant image distorts the geometric features of the input image. Conversely, fixing higher DCNN layers modifies input images to have high-level features stored within the DCNN imposed on them, resulting in hallucination-style images. Examples of these images can be found in Figure 6.

The original deep dream algorithm had to be modified in order to produce coherent and compelling hallucinatory style videos. The original algorithm was developed to process only static images (Mordvinstev et al., 2015). However, applying deep dream to every frame independently prevents the hallucinatory patterns and features from corresponding over time, such that the resulting videos are obfuscated. Alpha blending ensures this issue can be overcome. Specifically, the first frame of the video, after it has been deep dreamed, is blended with the next, unprocessed frame. Next, the deep dream algorithm is applied to the resultant, blended image, such that some of the hallucinatory patterns of the first frame are integrated into the second, and so on. The level

of blending can also be altered: in this study we simply used a ratio of 0.5. This process ensures that the hallucinatory content of the frames remains relatively consistent over time.

Different videos were used for each of the experimental blocks. Each video was processed with different deep dream parameters such that the videos display 'hallucinations' ranging from low-level geometric distortions to high-level feature representations. The various parameters used are displayed in Table 1. The unprocessed video of the lab was deep dreamed 5 separate times for each layer, such that there were a variety of patterns and 'hallucinations' visible throughout the experiment, thus preventing familiarity effects in participants.

Layer	**Octaves**	**Number of Iterations**
conv2d1_pre_relu/conv	4	3
mixed3a_pool_reduce_pre_relu/conv	5	3
Mixed4c	6	3
Mixed5a	6	4

Table 1: The different deep dream parameters used to create the videos that were used within the behavioural experiment. Static images taken from these videos are displayed in Figure 6.

3.2.5 Superstitious Perception Stimuli

100 (50cm x 50cm) Gaussian white noise images were created using MATLAB (MathWorks, 2017). They were presented within the head mounted display (HMD), fixed to the center of participants' field of view, such that wherever participants moved during the Superstitious Perception task, they were always looking directly at the noisy image.

3.2.6 Vividness of Visual Imagery Questionnaire (VVIQ)

The VVIQ has been used to determine the vividness of subjects' visual imagery (Marks, 1973). It is performed by asking participants to imagine different images in their mind, with eyes open and closed, and then to rate the vividness of the image on a five point scale, ranging from 'perfectly clear and as vivid as normal vision' to 'no image at all'. The final score can be used as a quantitative measure of subjects' vividness of visual imagination, in other words their ability to form images within 'the mind's eye'.

3.2.7 Post-Experiment Questionnaire

A post-experiment questionnaire was used to establish participants' subjective ratings of the experience. The questionnaire included ratings of presence (To what extent did you believe you were in the real world, as opposed to a virtual world?) and body shifting (To what extent did you feel your body position shift from your physical body?) on a 10 point scale. Furthemore, participants

Figure 6: Still images taken from the different videos used in the behavioural experiment. Videos processed with a lower DCNN layers show local geometric distortion, whilst the later videos have high-level representations imposed on them. The parameters used to create these videos are shown in Table 1. *Top Left*: Original Video, *Top Right*: conv2d1_pre_relu/conv, *Center Left*: mixed3a_pool_reduce_pre_relu/conv, *Center Right*: Mixed4c, *Bottom*: Mixed5a.

were asked whether they felt they saw more letters or numbers in the noisy images before or after the experiment.

3.3 Participants

25 participants completed the experiment (Mean Age= 23, SD= 3.52, 11 Female). Participants provided informed consent and received £7 after completing the experiment. Participants were excluded from the study if they had experienced clinical hallucinations in the past or if their eyesight was not 20/20 or corrected to 20/20. The University of Sussex Research Ethics Committee approved all experimental procedures.

3.4 Experimental Design

The experiment was undertaken in the Virtual Reality lab at the University of Sussex. After being fitted with the HMD, participants were instructed that a certain proportion of the superstitious perception stimuli contained hidden letters or numbers. Participants were then asked to report the number of images they believed to have the hidden letter within them, in a manner similar to the task utilised in previous investigations on superstitious perception (Gosselin, 2003). Participants reported seeing a hidden letter or number by pressing the right arrow key, and reported seeing nothing by pressing the left arrow key.

Next, participants were shown a live feed of the lab in which the experiment was taking place. After participants had had a chance to explore the feed through the HMD ($\sim$1 minute), participants were either instructed that the video they would be seeing was pre-recorded (NONSR condition; $N = 10$, Mean Age = 23, SD = 4.82) or that the live feed would continue (SR condition; $N = 15$, Mean Age = 23, SD = 3.55). All subjects, however, viewed pre-recorded videos. Subsequently, the experimenter switched the input such that participants viewed the pre-recorded video of the lab. If subjects were in the SR group (i.e. they were told they were viewing a live feed), the switch between live feed and pre-recorded video was made whilst participants moved their heads, as this technique has been shown to be effective at preventing subjects from noticing such changes (Suzuki et al., 2012). The camera providing the live feed was situated in the same location as when the video was filmed, such that participants were unable to detect when the change of input had occurred. The experimenter informed only those in the NONSR group when the switch to video was happening.

Once participants were viewing the pre-recorded video, they were instructed to report (via button press) whenever they noticed a change or manipulation to the visual input they were receiving. Each trial started with the unprocessed video of the lab, with the deep dreamed counterpart gradually emerging on top, such that by the end of the trial the video was solely the deep dreamed video. Each trial lasted for 30 seconds. The deep dream video was programmed to emerge via alpha blending, whereby the transparency of the original video increased linearly as the transparency of the deep dreamed video decreased linearly. To prevent participants from realizing the exact time

of onset of the deep dream video, we varied the time at which the blending process began, within a range of 1 to 10 seconds. Regardless of when the participant noticed the emergence of the deep dream, the video played until the end, such that all participants viewed the same amount of 'hallucinatory' content. The order of the videos (i.e. the layers of the network with which the videos had been produced) was randomized, as was the order of the different videos made within each layer.

After each trial, visual input went black for one second so that the participant was aware the trial was over, before the unprocessed video was displayed again at the start of the next trial. Each block consisted of 20 trials and there were 5 blocks throughout the experiment. The design of each trial is illustrated in Figure Seven.

After the last block, participants were asked to repeat the superstitious perception task they had performed at the start of the experiment. Finally, they completed the VVIQ and the post-experiment questionnaire before being debriefed.

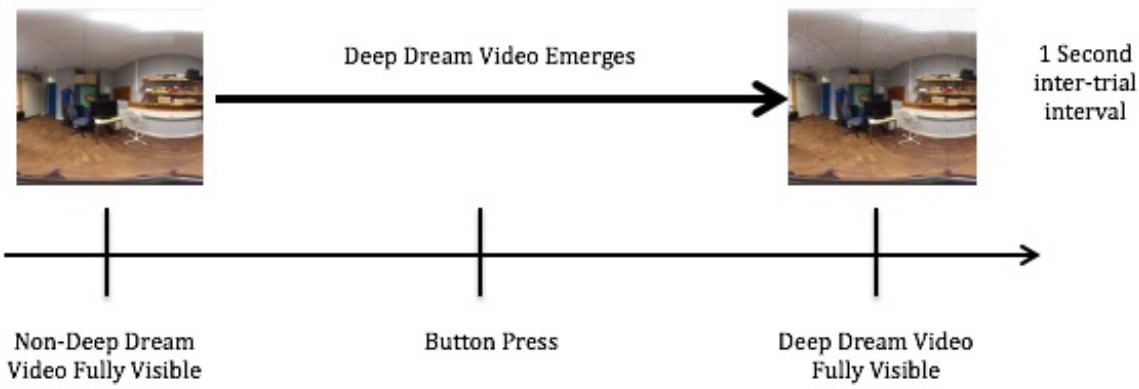

Figure 7: Experiment task structure. Participants initially view the unaltered 360-degree video of the VR lab. Over the course of 30 seconds, the deep dreamed video emerges on top of the original video, such that the hallucinatory patterns begin to emerge over time. Participants are instructed to press a button when they perceive any changes or modifications to the videos.

4 Results

4.1 Reaction Times Between SR and NONSR Conditions

An Independent Samples Welch's t-test showed that individuals in the SR group were no quicker to detect changes in the detection task (M = 8.97s, SE = 0.41s) than those in the NONSR group (M = 9.32, SE = 0.51), t(19.13) = -0.52, p = .60 This is illustrated in Figure 8.

Bayes factors were also calculated to assess the validity of evidence for a given effect. Bayesian statistics offers an alternative approach to analysing data and, unlike classical statistics, is based

on pre-existing knowledge, allowing for firm conclusions as to the validity of one's hypotheses and also indicates when more data is needed. As such, Bayesian analysis has been deemed by some to be a more powerful method of interpreting data (Dienes, 2011, 2014).

The Bayes Factor is the amount by which the data is best explained by the alternative hypothesis over the null hypothesis. As such, predicted effect sizes for the alternative hypotheses are required. Usually such predictions are estimated from previous work, however due to the unpredictable size and direction of the virtual hands' effects, a normal distribution was used to calculate the Bayes factors in these analyses.

A Bayes factor (BF_{10}) above 3 demonstrates substantial evidence against the null hypothesis, whereas $BF_{10} < 1/3$ offers support for the null hypothesis (Jeffreys, 1939, in Parkinson et al., 2016). Bayes factors in between 1/3 and 3 suggest that the data is insensitive and is unable to distinguish between the null and alternative hypothesis, thus suggesting more participants should be tested (Dienes, 2011).

A JZS Bayes Factor Independent Sample T-Test (Love et al., 2015) with default prior scales indicated that insufficient data had been collected to form a firm conclusion either for or against the hypothesis that utilising SR would reduce participants' reaction times ($BF_{10} = 0.41$).

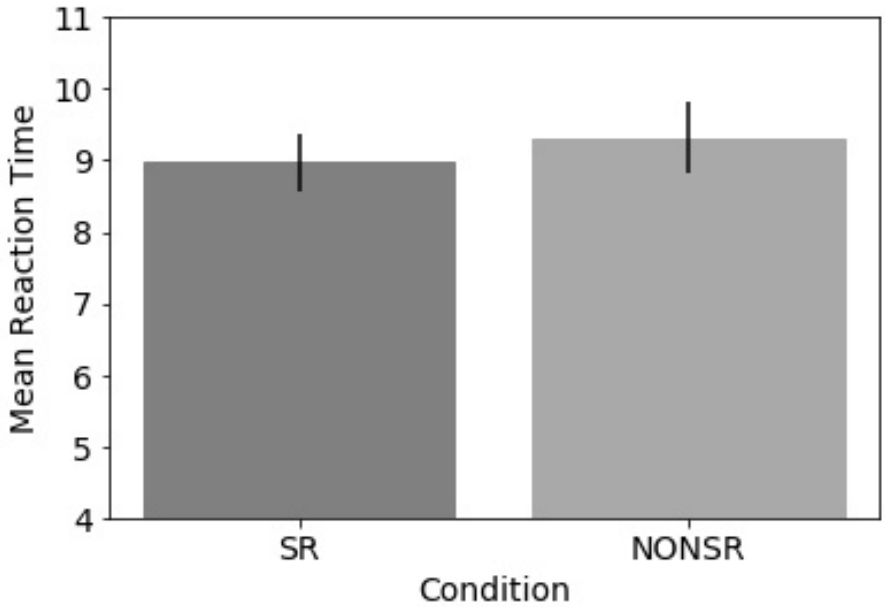

Figure 8: There was no significant difference in the time taken to detect the emergence of hallucinatory patterns between participants in the SR condition and those in the NONSR condition. Error Bars indicate Standard Errors.

4.2 Superstitious Perception Between SR and NONSR Conditions

Participants superstitious perception scores were calculated by finding the difference in 'Yes' responses after the experiment as compared to before the experiment took place. An Independent

Samples Welch's t-test revealed no significant difference between participants' superstitious perception when they were in the SR group (M= 1.6, SE= 2.34) or the NONSR group (M= −0.90, SE= 2.81), t(19.71) = 0.68, p= .502. This is illustrated in Figure 9.

A JZS Bayes Factor Independent Sample T-Test (Love et al., 2015) with default prior scales suggested that insufficient data had been collected either support or refute the hypothesis that utilising SR would increase participants' superstitious perception (BF_{10} = 0.44).

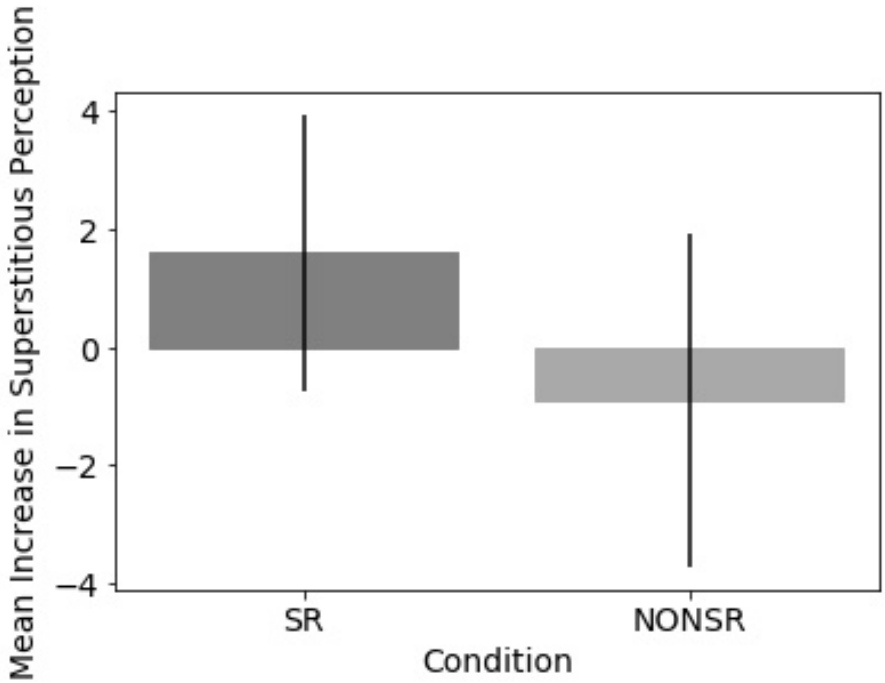

Figure 9: Participants in the SR condition did not increase their superstitious perception significantly more than those in the Non-SR group following immersion in the Hallucination Machine. Error Bars indicate Standard Errors.

4.3 Superstitious Perception After the Hallucination Machine

A two-way mixed ANOVA was conducted to compare the main effects of SR condition and time point of superstitious perception task and the interaction between these two variables on participants' superstitious perception scores. SR condition was the between subject factor and consisted of two levels: SR and Non-SR. Time point of the superstitious perception task was the within subject factor and also consisted of two levels: pre and post experiment. The main effect for SR condition yielded an F-Ratio of F(1,23) = 0.013, p= 0.91, whilst the main effect for time point of superstitious perception score returned an F-Ratio of F(1,23) = 0.036, p= 0.85. The interaction effect was also insignificant F(1,23) = 0.464, p= 0.50.

A JZS Bayes Repeated Measures ANOVA (Love et al., 2015) with default prior scales, performed on the same data returned a BF_{10} of 0.29 for time point of superstitious perception task,

supporting the conclusion that the time at which the superstitious perception task took place in relation to the rest of the experiment had no effect on superstitious perception scores. The same analysis yielded a BF_{10} of 0.66 for SR group, suggesting more data should be collected before any firm conclusion can be made regarding the effect of SR group on superstitious perception scores. Combining these factors, with and without the interaction, returned a BF_{10} of 0.20 and 0.09 respectively, offering strong indication of a non-significant interaction between SR condition and time point of superstitious perception task. There is thus substantive evidence that immersion in the Hallucination Machine does not impact superstitious perception, no matter whether individuals experience the Hallucination Machine with SR or not.

4.4 Subjective Reports of Superstitious Perception

Participants were asked to report whether they believed they saw more patterns in the noisy images either before or after immersion in the Hallucination Machine. In line with the objective measurements of superstitious perception, the majority of participants in the SR condition perceived there to be the same amount of illusory patterns both before and after the experiment. In contrast, however, more participants in the NONSR condition believed they saw more patterns in the noisy images after the experiment than before. These responses are illustrated in Figure 10.

4.5 Multivariable Regression

A multiple linear regression was calculated to predict participants' reaction times based on their VVIQ scores and superstitious perception scores. A non-significant regression was found ($F(2, 22) = 2.41, p = 0.11$) with an R^2 of 0.18. Only the VVIQ scores were significant predictors of participants' Reaction Times ($\beta = -0.041, p = .045$), whilst Superstitious Perception Scores were not ($\beta = 0.0076, p = .464$). In sum, the vividness with which participants saw mental imagery reliably predicted the time it took them to detect the emergence of hallucinatory patterns in the Hallucination Machine. Specifically, the more vivid their imagery, the longer it took them to detect changes.

4.6 Sense of Presence

In the post-experiment questionnaire, participants in the SR condition indicated they believed they were viewing a live feed, with an average rating of 7.13 out of 10 in response to the question: *'On a scale of 1-10, to what extent did you believe you were watching a live feed?'*

In an attempt to measure participants' sense of presence, they were asked: *'On a scale of 1-10, to what extent did you believe you were in the real world, rather than a virtual world?'* An Independent Samples Welch's t-test revealed no significant difference between participants' sense of

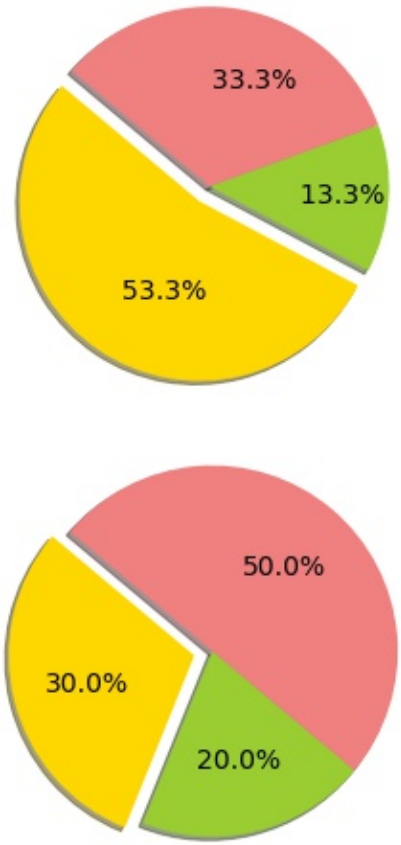

Figure 10: Pie charts showing the distribution of participants who believed they saw more patterns after the experiment, rather than before. *Above*: SR condition. *Below*: NONSR condition. *Red*: Perceived more patterns after the experiment; *Green*: Before experiment; *Yellow*: Equal.

presence when they were in the SR group (M= 5.733, SE= 0.77) or the NONSR group (M= 4.40, SE= 0.92), t(19.90) = −1.106, p= .282. This is illustrated in Figure 11.

A JZS Bayes Factor Independent Sample T-Test (Love et al., 2015) with default prior scales suggested that insufficient data had been collected either support or refute the hypothesis that utilising SR would increase participants' sense of presence ($BF_{10} = 0.58$).

5 Discussion

In brief, participants who underwent substitutional reality techniques, rather than just virtual reality, were no faster when detecting the emergence of hallucinatory patterns within the HMD. In line with this, subjective reports of participants' sense of presence did not differ between the groups. Secondly, viewing hallucinatory patterns within the HMD was found to have no effect on superstitious perception across participants; nor was it found to affect the SR group's superstitious perception more than the NONSR group's. Finally, VVIQ scores were found to correlate with participants' reaction times in the detection task, in such a way that higher ratings of vividness of

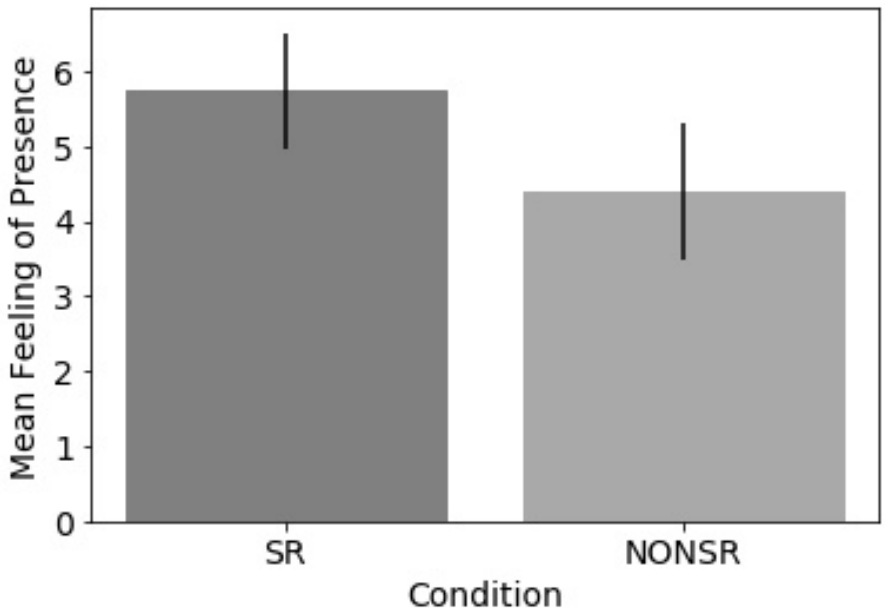

Figure 11: There was no difference between participants' responses to the question: '*On a scale of 1-10, to what extent did you believe you were in the real world, rather than a virtual world?* Error Bars indicate Standard Errors.

visual imagery predicted slower reaction times.

The observation that higher ratings of vividness of visual imagery in the VVIQ were associated with slower reaction times in the detection task was aligned with our hypothesis. This result is perhaps unsurprising when reasoning that individuals with vivid mental imagery will likely be habituated to the existence of vivid patterns in their 'mind's eye' and that this will therefore 'translate' to normal vision, such that the ability to detect the slow onset of patterns will be slightly weaker in such individuals. In contrast, those who have poor visual imagery can therefore detect changes sooner since the existence of vivid patterns in vision is a severe departure from their normal perception.

Despite this rationale, this result is in conflict with previous research in vividness of visual imagery, and the VVIQ specifically. Firstly, scores on the VVIQ have rarely been found to correlate with objective behavioural measures in any cognitive domain other than visual memory (Hilgard, 1981), defined as the ability to recall images from previous perceptual experiences. This statement is supported by a meta-analysis of VVIQ studies, which found no relationship between VVIQ scores and tests of cognitive abilities that would be expected to be influenced by mental imagery, such as mental rotation, spatial transformation or visual creativity (McKelvie, 1995). Indeed, the only domain found to have a significant relationship with VVIQ scores was memory. There is a large body of literature linking the VVIQ with memory, for example participants with high scores on the VVIQ were found to falsely recognise distractor words in a memory task, alongside being more likely to add erroneous words in a recall task (Winograd et al., 1998). Similarly, high imagery individuals

have been shown to be able to recall significantly more sensory details of previously seen items, as well as unseen items that have been described to them (Keogh and Markham, 1998). Taken together, these studies have been used to conclude that the VVIQ does not accurately measure vividness of visual imagery, but rather is a specific measure of visual memory (Slee, 1995). It is clear that our findings are in conflict with this view, as our simple change detection task does not rely on visual memory, but rather is a measure of the ability to perceive vivid patterns during natural vision. As such, the view that the VVIQ is merely a measure of visual memory should be doubted, and the further investigation of VVIQ scores with theoretically motivated behavioural measures should be encouraged, in order to ascertain the impact vivid mental imagery has on various cognitive processes.

Alternative discussions of the VVIQ have doubted the questionnaire's ability to measure anything related to visual imagery whatsoever (Rodway et al., 2006). For instance, it was found that confident individuals gave higher ratings of vividness on the VVIQ than less confident participants, but only if they displayed worse performance on a visual memory task (McKelvie, 1994). This has been used to support the notion that confident but inaccurate people are more likely to report vivid imagery during the VVIQ (Rodway et al., 2006), rather than simply those with vivid mental imagery. In this way, it may be hard to deduce whether the VVIQ is an accurate measure of vividness of visual imagery or some other cognitive factor. Furthermore, the domain of visual imagery is obviously multidimensional: comprising of multiple subprocesses (Farah, 1984; Dean and Morris, 2003), and, as such, the VVIQ may measure other dimensions of imagery such as clarity of image or controllability, rather than just vividness. Indeed, a componential analysis of the questionnaire described how the VVIQ fails to measure a large portion of the subcomponents associated with mental imagery (Kosslyn et al., 1984). Perhaps most importantly, the VVIQ may not offer an accessible way to communicate vividness of mental images. For instance, the top of the scale in the VVIQ reads: 'perfectly clear and as vivid as normal vision.' After consideration, it is hard to discern exactly what this means: is it possible to compare across two disparate perceptual experiences? Whilst there may be similarities, to compare clarity and vividness across the domains of normal and internal vision is perhaps as impervious to introspection as comparing clarity and vividness between sight and hearing. As such, the VVIQ may lack specificity in measuring vividness of imagery, alongside failing to enable effective communication of vividness of imagery.

Such shortcomings may help explain the relationship between VVIQ scores and poor detection ability. Perhaps, if the VVIQ is a measure of some other cognitive factor, such as controllability of visual images, participants with greater abilities within this domain will have more confidence in the belief that they have control over their visual scene and will therefore be later to detect external manipulations to their visual scene.

Overall, the finding that high scorers on the VVIQ were slower to detect the emergence of hallucinatory patterns in the Hallucination Machine can be used to dispute the suggestion that the VVIQ is purely a measure of visual memory. However, it is perhaps more likely that the VVIQ measures some other cognitive domain related to vividness of visual imagery, and that performance in our detection task relies on such a factor more so than vividness of visual imagery. It should be noted, however, the small sample size with which we conducted this study. Due to this small sample, statistical power for multilevel regression analysis is very low, and so any conclusions drawn from this study should be considered with caution. In any case, it would be beneficial to the field of visual imagery studies if further work focused on the VVIQ and its relation to behavioural measures of gradual change detection, in the attempt to identify exactly what the VVIQ is measuring.

Our results show no visible effect of hallucinatory visual stimulation on superstitious perception; calling our hypothesis that such visual stimuli would reduce the precision of bottom-up sensory input whilst increasing that of stored representations. It could be argued that the failure to elicit an effect on superstitious perception was due to the lack of fidelity in the visual representation of the lab within the HMD, since the video watched by participants contained visibly less detail than natural vision, therefore preventing immersion in the VE. However, this is unlikely to have impacted our results, since it has been found that visual detail is not as important as other parameters, such as head tracking and frame rates, in instilling a sense of presence during VR experiences (Sanchez-Vives and Slater, 2005). For example, participants who viewed a VE at differing levels of visual realism (from line drawings to photo-realism) showed no significant difference in heart rate when they reached a virtual precipice, despite all groups showing an increase in heart rates (Zimmons and Panter, 2003). Moreover, there was no difference in presence between the groups. In another experiment, performance in a task was equivalent across individuals who were either situated in a lab or a virtual simulation of the same lab (Usoh et al., 2000). Given the above, and the fact that the head tracking and frame rate in our Hallucination Machine are in real time, it seems unlikely lack of visual resolution prevented a sense of presence such that precision of priors was unaltered.

It is more likely the fact that the superstitious perception paradigm utilised was insensitive to any effect our set up had on precision weightings. In the study that first introduced the paradigm, participants viewed a far greater number of noisy image stimuli over a period of a fortnight (Gosselin, 2003), rather than the mere 100 our participants viewed. With the time restrictions we faced, using such a large number of sessions was not possible, but the use of fewer images was justified based on an alternative study using the same paradigm, which used only 24 images to show an effect of losing control on superstitious perception (Whitson, 2008). However, in this experiment, the authors used 12 images that did actually contain hidden patterns. The procedures used were therefore not identical, and as such, this could explain why our methods were unable to uncover

any effect of reducing the reliability of visual input on precision weightings. Further work with the Hallucination Machine, using more refined measures of precision weightings, are thus necessary before any conclusions can be made as to the effect of hallucinatory visual stimulation on the balance of top-down vs. bottom-up inputs during perception.

In contrast to our hypotheses, participants who believed they were viewing a live feed were no faster at detecting the emergence of patterns in the Hallucination Machine than those who knew they were watching pre-recorded videos. Our hypothesis was based on the assumption that the positive impact a greater sense of presence has on memory (Dinh et al., 1999; Lin et al., 2002; Mania and Chalmers, 2001) was due to more detailed perception with higher feelings of presence. However, this may not be the reason for the association between presence and memory. Indeed, an improved memory may arise for a number of reasons, such as greater emotional impact with more presence (Liebert, 2007; Kensinger and Corkin, 2003) or a greater propensity to reflect on the experience, leading to greater rehearsal of the scene (Craik and Lockhart, 1972). Neither of these potential explanations would necessarily cause participants to detect changes quicker than normal. Taking this into account, it is perhaps unsurprising that we found no difference in reaction times between conditions.

In line with this negative finding, we found no difference in participants' subjective ratings of presence. Whilst we predicted the SR methodology would improve participants' sense of presence, there are flaws within the VE that may have prevented a strong sense of presence being elicited. Most notably is the fact that participants are unable to view their own body in the VE. Indeed, bodily interaction with a VE is one of the key factors, alongside frame rate and head tracking, that maintains a robust sense of presence in participants (Sanches-Vives and Slater, 2005). As such, the fact that participants cannot see themselves or manipulate objects in the VE, will likely impact the extent to which participants feel present within the VE.

It may be possible to overcome this by asking participants not to look down or to not move their arms, but this is impractical in experiments that last a considerable amount of time. A more realistic, yet technically more demanding option, would be to overlay the kind of virtual hands used in a variety of VR experiments (Suzuki et al., 2013; Slater et al., 2008; Yuan and Steed, 2010), to the panoramic video. By using hand tracking hardware, the virtual hands' movements could be made to be congruent with the movement of the participant's actual hands, enabling some degree of interaction with the VE. Future work with SR should implement these ideas and compare the effects they have on the level of presence evoked.

SR methodology has, however, been shown to be resilient in maintaining a sense of presence in individuals. As mentioned previously, participants who had been informed of how the system's mechanism were still unable to differentiate pre-recorded from live scenes, even becoming confused

when viewing a live scene with live conversations (Suzuki et al., 2012). This strongly suggests the SR methodology is capable of instilling a sense of presence, despite offering no way for participants to interact with the VE. However, in the original SR set-up, participants were shown the scene from the same perspective as their physical body. This is a major difference to the present set-up, wherein participants are explicitly aware that the scene they are viewing is taken from an alternative location to their physical body. This may therefore have weakened the ability of the SR methodology to instill a sense of presence in comparison to that in Suzuki et al.'s study (2012), due to the reduced sense of reality imposed by viewing the VE from a disembodied location. The fact that the scene was not filmed from a first person perspective did not affect our ability to deceive participants into believing they were viewing a live feed, however. Indeed, 9 out of 15 participants in the SR condition reported that they believed they were viewing a live feed, with an average rating of 7.13/10, underscoring our success at misleading participants into thinking pre-recorded videos were in fact live.

In any case, the lack of difference in reaction times observed between the SR and NONSR group is not necessarily due to a failure to evoke a greater sense of presence in the former; the lack of difference found between the groups in terms of presence was likely due to the way in which participants were probed to report this aspect of their experience. Previous research on presence has used questionnaires with multiple questions, with scales from 0 to 100 (Dinh et al., 1999) or 0 to 14 (Hendrix and Barfield, 1996). In this way, it is perhaps unsurprising that the single question we asked after the experiment failed to identify what is likely to have been a subtle, yet significant, difference in feelings of presence between groups.

Subsequent studies with the Hallucination Machine should thus use better methods of measuring sense of presence in participants, such as the aforementioned questionnaires. Objective measures of presence may also be beneficial for deducing the effect of SR methods on presence and cognition, such as measuring skin conductance responses to fearful stimuli within the VE, a measurement that would be greater for those with higher feelings of presence (Meehan et al., 2002). It is our prediction that the utilisation of these tools would reveal greater feelings of presence in SR conditions, and this would explain why those in the SR group in the present study showed faster reaction times than those in the NONSR group.

6 Conclusion

Overall, we have failed to show that the use of SR methods can cause significant differences in cognition in virtual environments. However, if the neuroscience of consciousness is to continue to utilise VR to investigate aspects of cognition such as embodiment (Blanke and Metzinger,

2009), time perception (Suzuki et al., 2017) or memory (Plancher et al., 2010), it is imperative that the conscious experience individuals have within the VR set up is as similar to 'normal' waking consciousness as possible. Indeed, if the sense of presence is a structural component of consciousness (Seth et al., 2012), to experiment whilst participants feel significantly less presence than they do during natural experiences will not shine as much light on the underpinnings of natural experience as could be possible with SR methodologies. Indeed, if presence 'is the transportation of consciousness into an alternative, virtual reality' (Sanchez-Vives and Slater, 2005), then presence can be seen as consciousness within that VE. It thus makes sense to endeavour to improve presence in VEs, to ensure conclusions drawn from VR research about natural consciousness are not misled by what is actually research into a qualitatively different experiential state. In line with this, the present paper has provided an experimental procedure with which to explore methods that may improve feelings of presence in VEs, alongside analysing any subsequent effects on cognition.

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